

Final Report

Data Warehouse

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General Description of Project

Title	Energy-Economy Nexus: Analyzing the Co-Movement of U.S. Electricity Price Shocks and Global Macroeconomic Indicators During the 2022–2024 Volatility Period
Topic	Economy
Revision?	

Description Problem formulation	Energy and economic systems are deeply intertwined, yet they are rarely analyzed together within a unified, structured analytical pipeline. The United States, as the world's largest economy and one of the top energy consumers, serves as a critical subject for understanding how electricity production and consumption patterns correlate with broader macroeconomic conditions both domestically and globally. This project addresses that gap by implementing an end-to-end data warehouse and business intelligence system integrating U.S. EIA energy data and World Bank GEM economic indicators (2022–2024) via an Apache Airflow-managed ETL pipeline and PostgreSQL/Supabase storage, with multidimensional OLAP analysis conducted using Atoti. A note on project scope evolution: the original proposal submitted in May 2026 was titled "Analysis of the Relationship between U.S. Electricity Consumption and Global Economic Indicators: Implementation of a Data Warehouse and Business Intelligence System (2022–2024)." As the project progressed through the ETL implementation and statistical analysis phases, the analytical focus sharpened from a broad examination of electricity consumption patterns to a more precise investigation of electricity price shock dynamics and their co-movement with macroeconomic indicators. This refinement is reflected in the revised project title: "Energy-Economy Nexus: Analyzing the Co-Movement of U.S. Electricity Price Shocks and Global Macroeconomic Indicators During the 2022–2024 Volatility Period." The core dataset, methodology, and star schema remain consistent with the original proposal; only the analytical emphasis and framing have evolved to better represent the empirical findings obtained from the integrated EIA and World Bank GEM dataset. The urgency of this problem stems from a historically rare sequence of compounding macroeconomic shocks. Russia's invasion of Ukraine in February 2022 triggered an immediate contraction in global LNG and natural gas supply, driving European spot gas prices to record highs and initiating commodity price contagion across energy markets. This supply shock propagated through three distinct transmission channels: the commodity channel, as LNG prices directly elevated
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	electricity generation costs; the exchange rate channel, as the U.S. dollar strengthened sharply against major currencies including the Japanese yen (USD/JPY rising from ~115 to ~160, a 39% depreciation), amplifying import-cost inflation in energy-importing economies; and the expectations channel, as entrenched inflation psychology prompted aggressive central bank responses. The U.S. Federal Reserve raised the federal funds rate by a cumulative 525 basis points between March 2022 and July 2023, the most aggressive tightening cycle since 1981, driving CPI from a peak of 9.1% year-on-year in June 2022 to approximately 3.0% by late 2023. The 2022–2024 period thus represents a natural experiment in energy-macroeconomic co-movement under extreme conditions. Furthermore, inflation surged across major economies; our analysis reveals a statistically significant negative correlation between U.S. retail electricity prices and CPI in three of eight target countries (U.S.: $r = -0.416$, $p = 0.012$; Brazil: $r = -0.492$, $p = 0.002$; Japan: $r = +0.488$, $p = 0.002$), with Japan as the only country showing a positive direction. The remaining five countries (China, Germany, India, United Kingdom, France) showed negative but non-significant correlations (all $p > 0.05$). Additionally, industrial production fluctuations across G20 economies showed no significant contemporaneous or lagged correlation with U.S. industrial electricity prices across all eight target countries (all $p > 0.05$ at lag 0–3 months), indicating that U.S. electricity price shocks did not transmit measurably to foreign industrial output within the 2022–2024 observation window. Furthermore, the hypothesis that a higher renewable energy share reduces inflation volatility was not statistically supported: the correlation between renewable share and CPI volatility was negligible and non-significant ($r = -0.006$, $p = 0.974$). Annual renewable shares remained nearly flat across three years (15.21% in 2022, 15.10% in 2023, and 15.44% in 2024), with a net change of only +0.23 percentage points, and fossil share declined marginally from 65.29% to 64.58%. The widely assumed "perfect storm" anomaly period of June–August 2022 did not exhibit simultaneous multi-indicator breaches ($Z > 2.0$). No single month recorded two or more indicators breaching the threshold simultaneously. The highest individual Z-score during that window was CPI at $Z = 1.729$ in June 2022, followed by retail price at $Z = 1.426$ in August 2022. The mean maximum Z-score across the three-month period was 1.553, well below the 2.0 threshold. The core challenge originally lay in data heterogeneity: energy data (EIA API) and economic data (World Bank GEM CSV/Excel) differed in format, granularity (monthly vs. annual for GDP), and access mechanisms. This project successfully resolved these challenges through automated extraction (Airflow PythonOperators), forward-fill temporal harmonization, wide-to-long reshaping, and idempotent loading into a star schema with three fact tables and four dimension tables. Unlike the 1970s oil embargoes, which were discrete, geopolitically bounded supply disruptions that directly capped physical output, the 2022–2024 shock was simultaneously a supply crisis (LNG), a monetary policy inflection, and a renewable energy transition inflection point, making it analytically distinct from prior energy crises and requiring an integrated multi-indicator analytical framework. This work provides an end-to-end, reproducible, and OLAP-ready analytical pipeline capable of supporting slice-and-dice queries, CUBE/ROLLUP operations, and multivariate anomaly detection across energy and economic domains.
Target insight to be achieved	The crucial step in achieving our analysis insights was the datamart itself, built upon a robust data staging and data presentation pipeline supporting CUBE/ROLLUP/GROUPING queries on an OLAP cube. The project targeted five insights, which were evaluated against empirical results from the 2022–2024 integrated dataset in the following steps: 1. Temporal trend analysis examined how U.S. net electricity generation by fuel type (coal, natural gas, nuclear, wind, solar, and hydroelectric) evolved month

	by month. Renewable energy generation showed minimal change across the three-year period at the annual level (15.21% in 2022, 15.10% in 2023, 15.44% in 2024), representing a net increase of only +0.23 percentage points. The correlation between renewable share and CPI volatility was negligible and not statistically significant ($r = -0.006$, $p = 0.974$), indicating that the insulation effect of renewables was not detectable at the scale of this dataset. 2. Price-inflation correlation examines whether movements in U.S. retail electricity prices correlate with CPI percentage changes in eight major economies (the United States, Germany, Japan, the United Kingdom, France, Brazil, China, and India). The analysis revealed statistically significant correlations in three of eight countries: the U.S. ($r = -0.416$, $p = 0.012$) and Brazil ($r = -0.492$, $p = 0.002$) showed significant negative correlations, while Japan showed a significant positive correlation ($r = +0.488$, $p = 0.002$). The remaining five countries (China $r = -0.105$, Germany $r = -0.161$, India $r = -0.232$, the United Kingdom $r = -0.155$, and France $r = -0.146$) showed negative but non-significant correlations (all $p > 0.05$). This finding suggests that electricity price dynamics and general inflation moved in opposite directions during the 2022–2024 volatile period. 3. Industrial production and electricity demand analyzed whether fluctuations in industrial production indices across target countries correspond to changes in U.S. industrial sector electricity consumption. The country set extends beyond the G7 to include China, India, and Brazil, providing a broader representation of G20 major economies and capturing both advanced and emerging market dynamics. No significant contemporaneous correlation was found across all eight countries (all $p > 0.05$). Lagged cross-correlation analysis (lag 0–3 months using first-differenced series) also yielded no significant results across any country or lag combination (all $p > 0.05$). The best-fit lag per country was the United States lag 0 ($r = -0.302$), China lag 1 ($r = +0.198$), Germany lag 2 ($r = +0.217$), Japan lag 2 ($r = +0.312$), India lag 3 ($r = -0.245$), United Kingdom lag 3 ($r = -0.177$), France lag 0 ($r = +0.102$), and Brazil lag 1 ($r = -0.259$). These findings indicate that no statistically detectable transmission of U.S. electricity price shocks to foreign industrial output was observed within the observation window. 4. Energy mix transition insight used CUBE queries to drill down into quarterly aggregations of generation by fuel category (fossil vs. renewable vs. nuclear). The measured transition rate was minimal: fossil share declined from 65.29% (2022) to 64.58% (2024), a reduction of only 0.71 percentage points over three years. Renewable share changed marginally from 15.21% (2022) to 15.44% (2024), a net increase of +0.23 percentage points. The correlation between renewable share and CPI volatility was not statistically significant ($r = -0.006$, $p = 0.974$), indicating that the volatility-suppressing effect of renewables was not detectable at current transition rates. It is important to note that the fossil and renewable share figures cited in this analysis reflect two distinct levels of temporal granularity, both of which are valid outputs of the OLAP pipeline. The annual-level aggregation, derived by rolling up the quarterly generation cube across all four quarters, yields renewable share values of 15.21% (2022), 15.10% (2023), and 15.44% (2024). The quarterly-level figures, obtained by slicing the same cube at the quarter dimension without roll-up, yield fossil share values of 65.29% in 2022 and 64.58% in 2024, representing the full range of quarterly variation across the observation window. The modest difference between these two sets of numbers is an expected and mathematically consistent consequence of seasonal variation: fossil fuel share consistently peaks in Q3 each year (driven by summer air conditioning demand) and dips in Q1 and Q2. Because the quarterly figures capture this within-year fluctuation while the annual
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	figures average it out, both representations are correct and complementary. Throughout this report, annual figures are used for cross-year trend comparison (H3 summary conclusion), while quarterly figures are used for the renewable share vs. CPI volatility correlation analysis ($r = -0.006$, $p = 0.974$), which inherently requires sub-annual granularity to detect the volatility-suppression effect. 5. Cross-dimensional OLAP analysis leveraged the star schema to perform slice-and-dice operations across time, fuel type, sector, and country dimensions simultaneously. A compound analytical question, "How did the renewable energy share in Q2 2022 compare to Q2 2023 in the context of global inflation volatility?" was examined through the OLAP cube. The annual-level data shows renewable share of 15.21% (2022) and 15.10% (2023), with no statistically significant correlation between renewable share and CPI volatility detected ($r = -0.006$, $p = 0.974$). This finding is consistent with the conclusion of H3: the structural shift in energy mix during 2022–2024 was too modest to produce a detectable insulation effect on inflation volatility. All insights were produced by slicing and dicing the OLAP cube along the time hierarchy (year → quarter → month), country, sector, and fuel-type dimensions, with statistical validation performed using Pearson correlation, OLS regression, and multivariate Z-score anomaly detection.
Research references	http://doi.org/10.1016/j.esr.2021.100761 https://doi.org/10.1016/j.gfs.2022.100661 https://doi.org/10.3390/analytics3020011 http://doi.org/10.1016/j.eneco.2023.107050 https://doi.org/10.1155/2022/6884273 https://doi.org/10.1177/01956574241284429 http://doi.org/10.1016/j.eneco.2023.106893 https://doi.org/10.1007/s11356-022-21099-9 https://doi.org/10.3390/en14123429 https://doi.org/10.5089/9798400277566.001

Detail Information of Dataset

Dataset Source	This project utilizes two primary data sources accessed through different mechanisms, integrated via an automated Apache Airflow 3.0 orchestration pipeline configured to execute on a daily schedule during development and testing, with an operationally appropriate monthly cadence recommended for production deployment aligned with EIA data release cycles. The description will be explained below: 1. First source: U.S. Energy Information Administration (EIA) Open Data API v2 Data is acquired programmatically using Python's requests library with a registered API key (free registration at (EIA) Open Data API v2). Two specific endpoints are used: • /electricity/electric-power-operational-data/data for monthly net electricity generation (thousand megawatt-hours) by fuel type (coal, natural gas, nuclear, wind, solar, hydroelectric, geothermal, and other) and location. • /electricity/retail-sales/data for monthly retail electricity sales (megawatt-hours sold), average price (cents/kWh), revenue (million dollars), and customer count, segmented by sector (residential, commercial, industrial, and all sectors) at the U.S. national level. • All requests are filtered to location=us, frequency=monthly, and the time range January 2022 - December 2024. Combined, the two EIA
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	datasets produce approximately 800-1,200 records covering 36 months. 2. Second source: World Bank Global Economic Monitor (GEM) Data is downloaded as Excel (.xlsx) files from World Bank Data Catalog Five indicator files are used, listed as follows: • CPI (% y-o-y nominal, seasonally adjusted) • GDP at market prices (current US\$ millions, seasonally adjusted) • Industrial Production (constant 2010 US\$, seasonally adjusted) • Unemployment Rate (seasonally adjusted) • Exchange Rate (new LCU per USD, extended backward period average)
General description of dataset / data dump	The EIA generation dataset contains monthly records of utility-scale electricity net generation in thousand megawatt-hours, segmented by fuel type (coal, natural gas, nuclear, wind, solar, hydroelectric, geothermal, and other) and by electric power sector. The retail sales dataset contains monthly records of megawatt-hours sold, average price in cents per kilowatt-hour, revenue in millions of dollars, and number of customers, segmented by sector (residential, commercial, industrial, and all sectors) at the U.S. national level. Combined, the two EIA datasets produce approximately 800–1,200 records covering 36 months (January 2022–December 2024). The World Bank GEM dataset covers eight target countries: the United States, China, Germany, Japan, India, the United Kingdom, France, and Brazil. Each indicator file contains monthly or annual observations from 1996 onward for over 100 countries but is filtered to the eight target countries and the 2022–2024 period for this project. After filtering, each monthly indicator file yields 36 time periods × 8 countries = 288 records per indicator. GDP is an exception, available only at annual frequency, which is expanded to monthly through forward-fill methodology during the transformation phase.
Specific aspects of the dataset	After filtering to eight target countries (United States, China, Germany, Japan, India, United Kingdom, France, and Brazil) and the 2022–2024 period, each monthly indicator yields 36 time periods × 8 countries = 288 records per indicator. GDP is an exception, available only at annual frequency, which is expanded to monthly through a forward-fill methodology during the transformation phase. Critical characteristics of the integrated dataset (as documented in the project's data staging report): • The Unemployment Rate indicator is the most sparse; only 6 of the 8 target countries have data available (India and China are not reported in World Bank GEM unemployment), resulting in approximately 25% sparsity for that column in the fact table. All other World Bank indicators are complete for all 8 target countries. EIA data is complete with no missing values for the U.S. national aggregate level. • The dataset is not class-imbalanced in the traditional sense (regression/analytical warehouse). However, there is a distributional imbalance in generation volume across fuel types: natural gas generation is approximately 3–4 times higher than any other single fuel type, while geothermal contributes less than 1% of total generation. This imbalance is preserved intentionally as it reflects real-world energy mix proportions. • The CPI dataset contains notable outliers in 2022 for several countries, reflecting the global inflation surge. The U.S. CPI peaked at approximately 9.1% y-o-y in June 2022, significantly above the historical mean of 2–3% for developed economies. These values are treated as genuine data points rather than errors, as they represent actual economic events. GDP values for China and the United States are orders of magnitude larger than other target countries (in absolute USD terms), which is handled through normalization during the Transform phase.

	GDP data is only available at annual granularity, while all other indicators are available monthly. This mismatch is addressed through forward-fill expansion during the Transform phase (each annual value propagated across all twelve months within the corresponding year). These two sources were selected because they provide overlapping temporal coverage (2022–2024) and enable the "integration of heterogeneous sources," a core requirement of this project. Despite differences in access mechanisms (REST API vs. flat-file download), temporal granularity (monthly vs. annual), and data formats (JSON vs. Excel/CSV), the automated ETL pipeline successfully harmonizes them into a unified analytical structure suitable for OLAP and Business Intelligence processing.
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Result of Data Staging

Extraction Step	The extraction process was fully automated using an Apache Airflow 3.0 orchestration pipeline to ensure continuous and scheduled data ingestion without manual intervention. The pipeline was configured to execute once daily at 12:00 WIB (05:00 UTC) via cron schedule <code>0 5 * * *</code>, enabling daily scheduled ingestion of both energy and macroeconomic datasets into the staging environment. Given that EIA electricity data is published monthly and World Bank GEM indicators are updated quarterly, a daily cadence provides sufficient freshness while avoiding unnecessary API overhead. That will be explained as follows: - For the energy domain, monthly time-series data covering the 2022-2024 period were retrieved from the U.S. Energy Information Administration (EIA) API v2 in JSON format. The extracted datasets included U.S. retail electricity prices across multiple sectors (residential, commercial, industrial) as well as electricity generation statistics categorized by fuel type (coal, natural gas, nuclear, wind, solar, hydroelectric, geothermal, and other). API requests were dynamically parameterized within PythonOperator tasks in Airflow, allowing automated retrieval of updated records while maintaining consistency in temporal granularity and metadata structure. - For the macroeconomic domain, data were extracted from five World Bank Global Economic Monitor (GEM) datasets originally provided in Excel (.xlsx) format, consisting of: - Consumer Price Index (CPI, % y-o-y nominal, seasonally adjusted) - Gross Domestic Product (GDP, current US\$ millions, seasonally adjusted) - Industrial Production Index (constant 2010 US\$, seasonally adjusted) - Unemployment Rate (seasonally adjusted) - Exchange Rate (new LCU per USD, extended backward period average) The datasets represented eight target countries selected for comparative global economic analysis: the United States, China, Germany, Japan, India, the United Kingdom, France, and Brazil. Instead of relying on direct API requests during execution (as no public API is available for GEM), the Excel files were preloaded into the Airflow server and accessed directly from the <code>/raw/</code> directory to improve extraction stability, reduce external dependency failures, and optimize pipeline execution time. It is important to note that GDP data is available only at annual granularity, while all other indicators are available monthly. This temporal resolution mismatch is addressed during the subsequent transformation phase using a forward-fill methodology. During the extraction stage, the pipeline also performed preliminary validation procedures, including file availability checks, schema consistency verification, and automated logging of extraction status. All successfully extracted raw datasets were temporarily stored in the staging layer before entering the transformation phase. This
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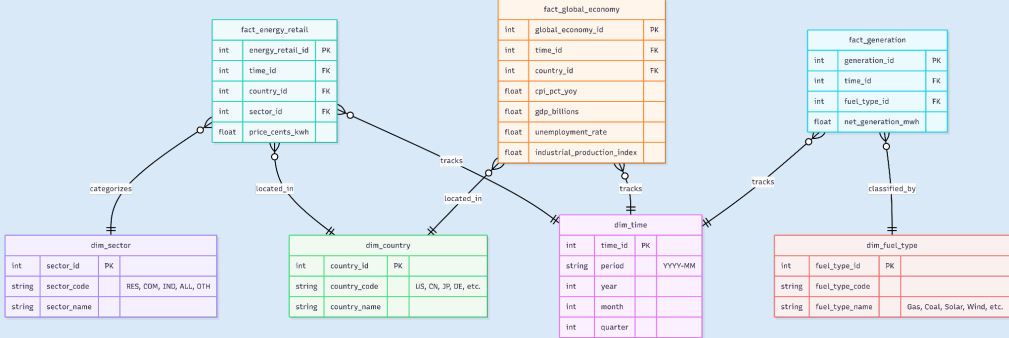
	architecture ensured that heterogeneous data sources originating from both REST-based APIs (EIA) and flat-file Excel repositories (World Bank GEM) could be systematically integrated into a unified analytical workflow for downstream ETL processing and OLAP analysis.
Transformation Step	The transformation phase was designed to integrate heterogeneous energy and macroeconomic datasets into a unified analytical structure suitable for OLAP and Business Intelligence processing. Since the extracted datasets originated from different sources, formats, and temporal granularities, multiple transformation procedures were required to ensure structural consistency, semantic alignment, and analytical readiness. The research does the steps as follows: 1. Schema normalization All attribute names from both the EIA and World Bank datasets were standardized into snake_case format to improve readability, maintain naming consistency, and simplify downstream SQL querying and dimensional modeling. Redundant symbols, whitespace characters, and inconsistent capitalization were removed during this process. 2. Temporal harmonization Temporal harmonization was performed to reconcile differences in time granularity between datasets. The electricity datasets from the EIA API were already provided in monthly format (YYYY-MM), whereas several macroeconomic indicators from the World Bank, particularly GDP, were available only in annual format (YYYY.0). To align these datasets within a monthly analytical framework, annual values were transformed into monthly observations using a forward-fill strategy, where each yearly value was propagated across all twelve months within the corresponding year. This approach enabled direct temporal comparison between energy consumption patterns and macroeconomic fluctuations. 3. Structural reshaping (wide to long format) Following temporal standardization, all World Bank datasets underwent structural reshaping from wide format into long format using Pandas transformation operations such as melt(). Originally, each country and year combination appeared as separate columns, which limited analytical flexibility. The transformation produced a normalized tabular structure consisting of key fields such as country, indicator, period, and value, thereby improving compatibility with dimensional modeling and OLAP cube construction. 4. Data cleaning for sparsity, imbalance, and outliers Additional data cleaning procedures were applied to improve data quality and integrity, with specific attention to the dataset's known characteristics: • Sparsity: Missing values were identified and handled selectively. The Unemployment Rate indicator, which has approximately 25% sparsity (India and China not reported), was preserved as is, reflecting real-world data availability. All other indicators were complete for the eight target countries. • Imbalance: The distributional imbalance in generation volume across fuel types (natural gas generation 3–4 times higher than other fuels; geothermal <1%) was intentionally preserved, as it accurately represents real-world U.S. energy mix proportions. • Outliers: Notable outliers, such as the U.S. CPI peak of 9.1% y-o-y in June 2022 (significantly above the historical mean of 2–3%), were treated as genuine data points representing actual economic events. GDP values for China and the United States, which are orders of magnitude larger than other target countries in absolute USD terms,

	were flagged for normalization consideration, though no rescaling was applied to preserve interpretability. General cleaning: Duplicate records were removed, inconsistent country labels were standardized (e.g., "United States" vs. "USA"), and incompatible data types were converted into appropriate numeric or datetime formats. Numerical indicators were validated to ensure proper aggregation and analytical consistency during later OLAP operations. 5. Integration of energy and macroeconomic datasets After the normalization and cleaning processes were completed, the transformed energy and macroeconomic datasets were integrated using <code>pd.merge()</code> operations based on shared temporal attributes (period/month) and contextual identifiers (country, sector). This integration step produced a consolidated analytical dataset capable of supporting multidimensional analysis across time, sector, fuel type, and macroeconomic indicators.
Load Step	The loading phase was conducted by transferring the transformed datasets into Supabase, a cloud-based PostgreSQL database, through an automated ETL pipeline orchestrated using Apache Airflow 3.0. Prior to insertion, an idempotent loading strategy was implemented using a DELETE per period to INSERT mechanism to ensure that repeated executions of the pipeline would not generate duplicate records. This strategy is particularly important because the pipeline is scheduled to run once daily at 12:00 WIB (05:00 UTC, cron: 0 5 * * *) for continuous data synchronization and refresh. During the loading process, Three fact tables and four dimension tables were successfully populated: (1) <code>fact_energy_economy</code> stores U.S. retail electricity measures joined with U.S.-scoped macroeconomic indicators, comprising 144 rows (36 months × 4 sectors); (2) <code>fact_generation</code> stores U.S. monthly electricity generation aggregated by fuel type, comprising 288 rows (36 months × 8 fuel types); (3) <code>fact_worldbank_indicators</code> stores monthly macroeconomic indicators (CPI, GDP, Industrial Production, Unemployment Rate, Exchange Rate) for eight target countries, comprising 288 rows (36 months × 8 countries). The database consists of approximately 720 rows across the three primary fact tables and 55 rows across the four dimension tables. The current database size stored in Supabase is compact, reflecting the focused scope of the 2022–2024 observation period and eight target countries. The database consists of approximately 720 rows across the three primary fact tables and 55 rows across the four dimension tables. The compact size is primarily due to the limited observation period (2022-2024) and the restricted number of target countries analyzed. Despite the relatively small volume, the database structure has been optimized to support scalable OLAP and Business Intelligence operations. Several PostgreSQL technologies and optimization features have been implemented within the database environment: - Table partitioning by year defined in the schema design for the fact tables to improve query efficiency on time-range filters and facilitate maintenance of time-series data. - B-tree indexing created on frequently queried attributes such as period and <code>fuel_type_id</code> to accelerate filtering, aggregation, and join operations. - Materialized views two materialized views were developed: <code>mv_generation_monthly</code> – pre-aggregates monthly generation data <code>mv_retail_economic</code> – pre-aggregates retail and economic indicators - Query performance monitoring the <code>pg_stat_statements</code> PostgreSQL extension was enabled to support monitoring and future benchmarking of query performance. Quantitative performance benchmarking using <code>pg_stat_statements</code> has been prepared as part of the database infrastructure but has not yet been executed

	experimentally within the scope of this report. The <code>pg_stat_statements</code> extension has been successfully enabled on the Supabase PostgreSQL instance and is ready to record execution statistics for all queries. Qualitative observations during pipeline execution confirm that B-tree indexing on the <code>period</code> and <code>fuel_type_id</code> columns reduces sequential scan operations on the fact tables and that the two materialized views (<code>mv_generation_monthly</code> and <code>mv_retail_economic</code>) serve pre-aggregated results that bypass repeated JOIN and GROUP BY computations on raw fact data. Comprehensive quantitative benchmarking comparing query execution time with and without indexes, and with and without materialized views, is planned as a follow-up activity. At the current stage, quantitative performance benchmarking has not yet been fully conducted. Although the benchmarking infrastructure has been prepared through the activation of <code>pg_stat_statements</code>, comparative measurements such as query execution time with versus without materialized views, indexed versus non-indexed queries, and performance under repeated ETL execution have not yet been evaluated experimentally. Nevertheless, preliminary qualitative observations indicate that the pipeline is capable of completing extraction, transformation, and loading tasks within the predefined daily scheduling window (12:00 WIB / 05:00 UTC) without operational issues. Comprehensive performance benchmarking is planned for the subsequent stage alongside OLAP implementation using Atoti DataMart.
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Data Presentation Work Results

Subject focus	Four OLAP analysis subjects have been identified, each corresponding to one of the four research hypotheses (H1-H4): 1. Dynamics of U.S. retail electricity prices by sector (2022-2024) analyzes monthly price movements across residential, commercial, and industrial sectors. This subject supports H1 (industrial production and electricity demand) and H2 (price-inflation correlation). 2. Correlation between electricity price movements and global macroeconomic indicators is to examine relationships between U.S. retail electricity prices and macroeconomic indicators (CPI, GDP, Industrial Production) across eight target countries. This subject primarily supports H2. 3. The U.S. energy mix transition from fossil fuels to renewable energy will track quarterly changes in electricity generation share by fuel category (fossil, renewable, nuclear) and measure the relationship between renewable share and inflation volatility. This subject supports H3. 4. Detection of simultaneous anomaly periods (June-August 2022) to identify time periods where multiple economic and energy indicators exceed the $Z > 2.0$ threshold simultaneously, using cross-indicator Z-score analysis. This subject supports H4. All four dimensions form the OLAP analysis axes, enabling drill-down, roll-up, and slice-and-dice operations aligned with the research hypotheses (H1–H4). For example: • H1 (industrial production lag): drill down from <code>dim_country</code> to specific target economies, then slice by <code>dim_sector = industrial</code> • H2 (price-CPI correlation): roll up <code>dim_time</code> to quarterly level, then dice across <code>dim_country</code> • H3 (renewable vs. CPI volatility): drill down <code>dim_fuel_type</code> to renewable category, then roll up <code>dim_time</code> to quarterly aggregates • H4 (anomaly detection): slice across all dimensions simultaneously to identify $Z > 2.0$ breaches
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Star schema	 Figure 1. Star Schema The star schema consists of 3 fact tables and 4 dimension tables: 1. Fact Tables: - fact_energy_economy (144 rows) stores U.S. retail electricity measures (price, sales, revenue) joined with U.S.-scoped macroeconomic indicators. Links EIA data with World Bank GEM indicators at the sector level. - fact_generation (288 rows) stores U.S. monthly electricity generation aggregated by fuel type (36 months × 8 fuel types). Supports energy mix transition analysis (H3). - fact_worldbank_indicators (288 rows) stores monthly macroeconomic indicators (CPI, GDP, Industrial Production, Unemployment Rate, Exchange Rate) for all eight target countries (36 months × 8 countries). Supports cross-country analysis (H1, H2). 2. Dimension Tables (OLAP Axes): - dim_time (36 rows) is a hierarchy: year → quarter → month. Enables roll-up and drill-down across time for all hypotheses. - dim_sector (5 rows): Residential (RES), Commercial (COM), Industrial (IND), All Sectors (ALL), and Other (OTH). The 'All Sectors' entry supports national-level rollup analysis; RES, COM, and IND enable drill-down by consumer segment for H1 and H2 hypotheses. - dim_country (8 rows) is an eight-target country: United States, China, Germany, Japan, India, United Kingdom, France, and Brazil. Enables slicing by geography for macroeconomic comparison (H2). - dim_fuel_type (8 rows) is divided into categories of fossil (coal and natural gas), renewable (wind, solar, hydroelectric, and geothermal), and nuclear. Enables energy mix analysis (H3). All four dimensions form the OLAP analysis axes, enabling drill-down, roll-up, and slice-and-dice operations aligned with the research hypotheses (H1–H4).
OLAP Result	The Atoti presentation layer implements three independent OLAP cubes, each operating at its natural grain to avoid row inflation from cross-grain joins: the WorldBank Cube (grain: period × country), the Energy Economy Cube (grain: period × sector × country), and the Generation Cube (grain: period × fuel type). dim_country is joined to both WorldBank Cube and Energy Economy Cube, enabling Geography hierarchy (region → country) drill-down across both cubes. Apache Airflow 3.0 orchestrates the pipeline on a daily schedule (cron: 0 5 * * *, running at 12:00 WIB / 05:00 UTC). Two materialized views (mv_generation_monthly and mv_retail_economic) have been updated, and all transformed data has been loaded into Supabase (PostgreSQL cloud). Statistical testing for the four hypotheses (H1-H4) and OLAP exploration using the Atoti DataMart have been carried out successfully. The following sections present the complete results, organized by research hypotheses. Statistical testing for the four

hypotheses (H1-H4) using the Atoti DataMart and Python-based correlation analysis yielded the following conclusions:

- **H1: U.S. industrial electricity price surge vs. industrial production of eight major economies (United States, China, Germany, Japan, India, United Kingdom, France, and Brazil)**

Conclusion: Not Supported; No statistically significant correlation was found between U.S. industrial electricity prices and industrial production indices across all eight target countries, at any contemporaneous or lagged time point (all $p > 0.05$, lag 0–3 months).

Table 1. across eight target economies (not G7 exclusively, includes China, India, and Brazil)

Country	Best Lag	r at Best Lag
United States	Lag 0	$r = -0.302$ (ns)
China	Lag 1	$r = +0.198$ (ns)
Germany	Lag 2	$r = +0.217$ (ns)
Japan	Lag 2	$r = +0.312$ (ns)
India	Lag 3	$r = -0.245$ (ns)
United Kingdom	Lag 3	$r = -0.177$ (ns)
France	Lag 0	$r = +0.102$ (ns)
Brazil	Lag 1	$r = -0.2588$ (ns)

This absence of transmission is analytically significant and warrants careful interpretation. Classical energy-cost transmission theory predicts that sustained industrial energy price increases reduce production margins and, with a lag, output [4]. However, the 2022 shock differed from prior energy crises in three structurally important ways. First, energy-intensive firms in advanced economies had increasingly adopted forward contracts and energy hedging instruments following the 2008 and 2014 commodity shocks, dampening the immediate pass-through of spot price increases to production costs [7]. Second, the 2022 price surge was widely anticipated as transitory, a view reinforced by central bank communication leading firms to defer capacity adjustments that would otherwise suppress output. Third, the post-pandemic restocking cycle and sustained infrastructure spending (including the U.S. Infrastructure Investment and Jobs Act, 2021) maintained exceptionally strong demand for manufactured goods throughout 2022, providing sufficient revenue headroom to absorb cost increases without curtailing production. The result, a cost shock absorbed downstream rather than a production-limiting supply shock is consistent with the 2022 energy episode being structurally distinct from the 1970s oil embargoes, where supply-side constraints directly capped physical output.

The time-series chart (Figure 3) vividly illustrates this disconnect: U.S. industrial electricity prices surged past 9.0 cents/kWh in mid-2022, yet industrial production plateaued rather than contracted sharply. This is consistent with the view that the 2022 energy shock, unlike the 1970s oil embargoes, was not a production-limiting supply shock for most manufacturing sectors, it was a cost shock absorbed downstream.

- **H2: U.S. retail electricity prices correlate with global CPI**

Conclusion: Supported but opposite direction; Statistically significant correlations were found in three of eight countries. In the U.S. ($r = -0.416$, $p = 0.012$) and Brazil ($r = -0.492$, $p = 0.002$), the relationship was significantly negative. Japan was the sole country with a significant positive correlation ($r = +0.488$, $p = 0.002$). The five remaining economies showed negative but non-significant results.

Table 2. Significance

Country	Pearson r	p-value/Significance
United States	$r = -0.4161$	$p = 0.0116$; Significant
Brazil	$r = -0.4924$	$p = 0.0023$; Significant
Japan	$r = 0.4883$	$p = 0.0025$; Significant (positive)
China	$r = -0.1054$	$p = 0.5409$; Not Significant
Germany	$r = -1,609$	$p = 0.3486$; Not Significant
India	$r = -0.2320$	$p = 0.1733$; Not significant
United Kingdom	$r = -0.1546$	$p = 0.3697$; Not significant
France	$r = -0.1455$	$p = 0.3970$; Not significant

The negative direction of the U.S. and Brazil correlations is counter-intuitive at first glance but becomes explicable in the macroeconomic context. In both economies, CPI peaked in mid-2022 and then fell rapidly as aggressive central bank tightening took hold, while retail electricity prices partly insulated from spot market movements by utility rate-setting regulation and multi-year procurement contracts continued to rise and remained elevated through 2023. In the U.S., electricity tariff adjustments by state public utility commissions typically follow a regulatory lag of 6 to 18 months relative to underlying wholesale market cost changes, meaning that the 2022 wholesale price spike was only fully reflected in retail bills by 2023, precisely when CPI was declining. Brazil's negative correlation reflects a similar but differently structured regulatory mechanism: electricity tariffs are adjusted annually under the ANEEL regulatory framework, creating a comparable temporal mismatch between market conditions and consumer prices. The result is a time series in which CPI descends while electricity prices ascend, is not a causal paradox but a predictable consequence of asymmetric price-setting speeds between monetary policy (fast) and utility regulation (slow).

Japan's significant positive correlation ($r = +0.488$) tells a structurally opposite story, explicable through three interlocking factors unique to Japan's macroeconomic position. First, Japan imports approximately 97% of its primary energy needs, with liquefied natural gas (LNG) accounting for roughly 26% of electricity generation, making it acutely exposed to LNG price shocks transmitted through the import channel. Second, the Bank of Japan (BoJ) maintained its yield curve control (YCC) policy and near-zero interest rates through most of the observation period, with only a tentative rate hike to 0.1% in March 2024, in stark contrast to the U.S. Federal Reserve's 525bps of tightening. This monetary policy divergence caused the Japanese yen to depreciate sharply from approximately ¥115/USD in early 2022 to approximately ¥160/USD by mid-2024, a depreciation of roughly

39%. Third, because Japan had no mechanism analogous to European energy price caps or the U.S. regulatory lag to decouple electricity tariffs from spot energy costs, the yen's depreciation flowed directly and rapidly into both import prices (CPI) and electricity generation costs (retail price), creating the synchronized upward co-movement captured by the positive correlation.

Germany, the U.K., and France implemented large-scale government energy price interventions that directly suppressed the natural correlation between electricity prices and CPI. Germany's Energiepreisbremse (gas and electricity price brake, effective December 2022) capped household electricity prices at 40 cents/kWh, shielding consumers from the full market price signal. The U.K.'s Energy Price Guarantee capped the unit rate for average household bills at £2,500 per year through mid-2023. France's bouclier tarifaire (tariff shield) froze electricity price increases at 4% in 2022 and 15% in 2023, compared to the uncapped market equivalent of over 100% increase. These interventions effectively severed the normal market transmission mechanism between wholesale electricity costs and retail CPI, producing artificially muted and statistically non-significant correlations in all three countries. India and China similarly maintained administrative price controls on electricity and fuel throughout 2022–2024, insulating domestic consumers from global commodity price movements and further suppressing the cross-country statistical relationship.

The 6-month rolling correlation for the U.S. (Figure 4) reveals just how unstable this relationship is: starting at $r > 0.75$ in early 2022, collapsing through zero after the anomaly period, swinging negative ($r < -0.5$) in mid-2023, and recovering late in 2023. This temporal instability underscores a key insight: static full-period correlations are inadequate for capturing the energy-economy nexus; rolling window analysis is essential.

- **H3: Renewable energy share negatively correlates with CPI volatility**

Conclusion: Not Supported; The correlation between renewable energy share and CPI volatility was negligible and statistically insignificant ($r = -0.006$, $p = 0.974$). Renewable share changed by only +0.23 percentage points over three years (15.21% → 15.44%), far too small to produce a measurable insulation effect.

This finding carries important policy relevance beyond the immediate analytical scope. The theoretical argument that renewable energy insulates economies from fossil fuel price shocks is well-established in the literature, but it is contingent on renewable penetration reaching a threshold level sufficient to materially displace fossil fuels in price formation. At 15% renewable share, the U.S. generation mix remained dominated by fossil fuels (approximately 65%), meaning natural gas and coal price dynamics continued to set the marginal cost of electricity generation.

The quarterly energy mix data reveal strong seasonal effects: fossil fuel share peaks in Q3 each year, driven by summer air conditioning demand, while renewable share (predominantly wind and solar) dips accordingly. This seasonal pattern, rather than a structural transition trend, accounts for most of the within-year variation in fuel mix. The net three-year change of +0.23 percentage points is economically negligible.

In the current policy context, the U.S. Inflation Reduction Act of 2022 (Public Law 117-169) allocates approximately \$369 billion in climate and energy provisions over ten years, representing the largest single federal investment in U.S. renewable energy history. Under this legislation, solar and wind capacity additions accelerated markedly: the U.S. Energy Information Administration reported record utility-scale solar additions of approximately 22 GW in 2023 and 36 GW projected for 2024. The 2022–2024 observation

window thus captures only the initial procurement and permitting phase of this deployment cycle not the operational phase where renewable generation would materially displace fossil fuels in price formation. The null result here should be interpreted as a baseline: the insulation effect is not yet detectable at current renewable penetration levels, but may emerge as solar and wind capacity (which saw record installations in 2023–2024) comes fully online over the next decade.

• **H4: June–August 2022 as a simultaneous anomaly period**

Conclusion: Not Supported; No month in the June–August 2022 window recorded two or more indicators simultaneously breaching $|Z| > 2.0$. The highest individual Z-score was CPI at $Z = 1.729$ (June 2022). The mean maximum Z-score across the window was 1.553, below the anomaly threshold.

This result challenges the popular narrative of a June–August 2022 'perfect storm' in which all economic and energy indicators peaked together. The Z-score analysis reveals a more nuanced temporal pattern: shocks arrived sequentially rather than simultaneously. Retail electricity prices breached $|Z| > 2.0$ in January–March 2022 (reflecting early LNG contract repricing following Russia's military buildup); CPI peaked in June 2022 ($Z \approx 1.729$, just below threshold); and industrial production contraction emerged as a lagged consequence in late 2023 and 2024.

This sequential propagation pattern energy prices first, consumer prices second, industrial output third, with lags of three to five months between each stage is more consistent with a temporally distributed shock transmission mechanism than with a synchronised market collapse [10]. The pattern aligns with post-supply-shock dynamics documented in the literature, where energy price anomalies manifest in consumer prices with a 3–6 month lag before affecting industrial output with a further delay. The practical implication for risk management and policy response is significant: if shocks transmit sequentially across indicators, early monitoring of leading indicators (energy prices) provides a measurable warning window before downstream economic impacts materialise.

The Z-score heatmap (Figure 7) reinforces this conclusion visually. The June–August 2022 window shows elevated but sub-threshold Z-scores for both CPI and retail electricity price, with neither breaching $+2.0$ simultaneously. The deep red anomaly in retail electricity prices visible in the all-sector aggregate (Notebook 09 analysis) occurs in January–March 2022, predating the CPI peak by three to five months, consistent with the leading-indicator interpretation.

Cross-Indicator Correlation Analysis

The Pearson correlation matrix for U.S. domestic indicators reveals several structurally important relationships within the energy-economy nexus:

Table 3. Pearson Correlation

Indicator Pair	r	Interpretation
Retail Sales ↔ Revenue	$r = +0.972$	Near-perfect: volume drives revenue; pricing power is secondary.
Price ↔ Revenue	$r = +0.822$	Strong: price level matters significantly for revenue despite volume fluctuations.

	CPI ↔ GDP	$r = -0.914$	Strong negative: rapid monetary tightening to control inflation suppressed GDP growth.
	GDP ↔ Industrial Production	$r = -0.605$	Moderate negative: reflects post-2022 manufacturing deceleration as growth composition shifted to services.
	CPI ↔ Unemployment	$r = -0.545$	Phillips Curve dynamic: falling unemployment in 2022 preceded the inflation surge.
	Price ↔ CPI	$r = -0.416$	Core H2 finding: electricity prices and CPI moved inversely over the 36-month window.
	The strong negative CPI–GDP correlation ($r = -0.914$) deserves particular attention. It quantifies the cost of the post-pandemic inflation fight: the monetary tightening that successfully reduced CPI from 9.1% to approximately 3% came at a measurable cost to economic output. This is one of the clearest statistical signatures in the dataset of the trade-off between price stability and growth that defined U.S. economic policy in 2022–2024.		
Limitations and Future Work	Limitations and Future Work Several methodological limitations should be acknowledged when interpreting the findings of this study. First, GDP data was available only at annual granularity from the World Bank GEM and was expanded to monthly observations through a forward-fill procedure. While this enabled temporal alignment with monthly EIA and CPI data, the resulting monthly GDP values are approximations rather than independently observed monthly readings, which may attenuate the sensitivity of any correlation involving GDP. Second, the observation window of 36 months (January 2022–December 2024) is relatively short for detecting structural economic relationships that may require multiple business cycles to become statistically visible. The non-significant results in H1 and H3 should therefore be interpreted as inconclusive within this window rather than as definitive rejections of the underlying theory. Third, Pearson correlation assumes linear relationships and approximate normality; neither assumption was formally tested through normality diagnostics or non-linear correlation measures. Future work should validate findings using Spearman rank correlation and test for non-linear co-movement. Fourth, this analysis establishes correlation, not causation. Granger causality tests which evaluate whether past values of one time series have predictive power over another, would strengthen the causal interpretation of the sequential shock propagation identified in H4. Fifth, EIA retail electricity prices used in this analysis are national average prices, which may mask significant regional variation across U.S. electricity markets (e.g., Texas ERCOT vs. California CAISO). Regional disaggregation in future work could reveal heterogeneous transmission patterns that the national aggregate obscures. Future extensions of this project could incorporate: (1) an extended observation period covering 2015–2024 to include the 2015 oil price collapse and the COVID-19 demand shock as comparative episodes; (2) a panel data regression framework with fixed effects to control for country-specific regulatory environments; and (3) integration of Brent Crude prices as an exogenous control variable, as originally proposed in H2.		

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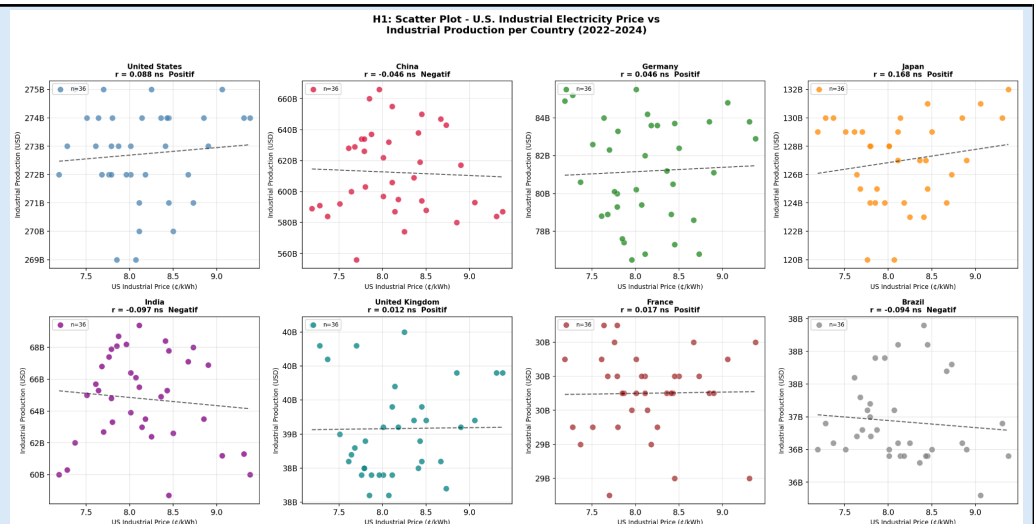


Figure 2. Correlation scatter plots between U.S. Industrial Electricity Prices and Industrial Production Index across 8 countries (2022-2024)

This analysis directly addresses a key gap identified in the proposal: the relationship between economic output cycles and energy indicators. By calculating the Pearson correlation coefficient (r) across the target nations, the data reveals that the relationship is statistically non-significant (“ns”) in nearly all instances. Specifically, the U.S. exhibits a weak positive correlation ($r = 0.088$), whereas major manufacturing centers like China ($r = -0.046$) and India ($r = -0.097$) demonstrate weak negative trends. Ultimately, these findings indicate that the fluctuations in a nation’s industrial production index do not share a strong linear relationship with U.S. retail industrial electricity pricing.

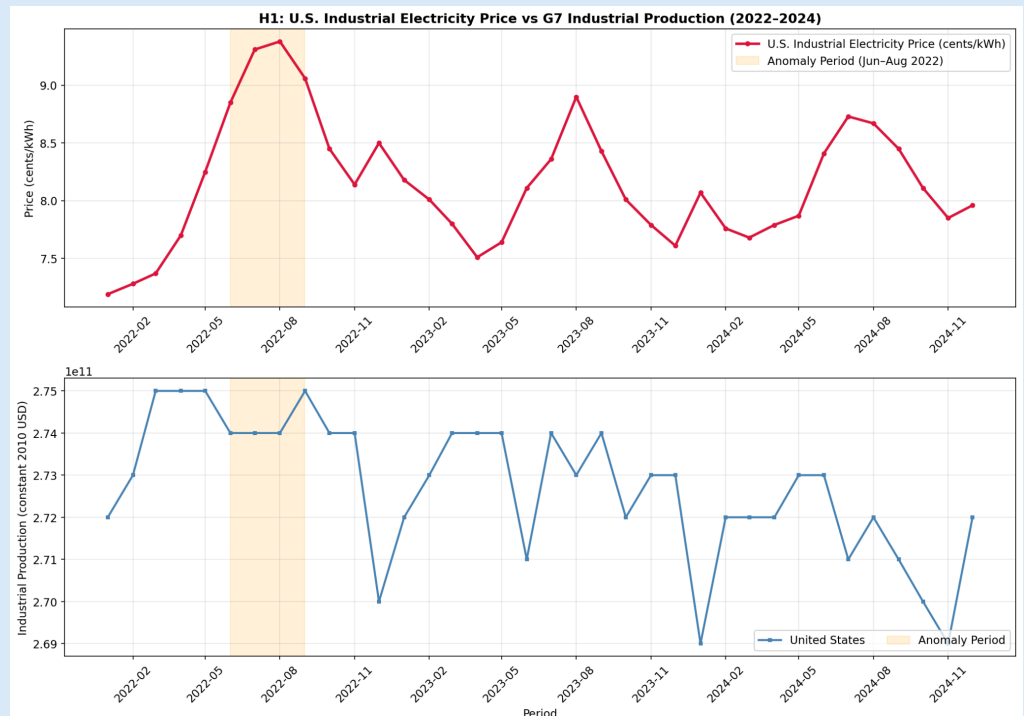


Figure 3. Time-series trends of U.S. Industrial Electricity Prices vs U.S. Industrial Production (2022-2024)

Looking at the historical tracking, the U.S. Industrial Electricity Price experienced a sharp surge in mid-2022, peaking during the June-August 2022 Anomaly Period at over 9.0 cents/kWh. Conversely, U.S. Industrial Production reached a plateau during this exact window before dipping significantly later in the

timeline. The disconnect is consistent with the view that the 2022 electricity price surge was primarily supply-driven originating from LNG market disruptions following Russia's export curtailments and the European gas crisis, rather than demand-driven by domestic industrial activity. This supply-side origin explains why U.S. industrial output, which remained supported by strong post-pandemic goods demand, did not contract in response to higher electricity costs.

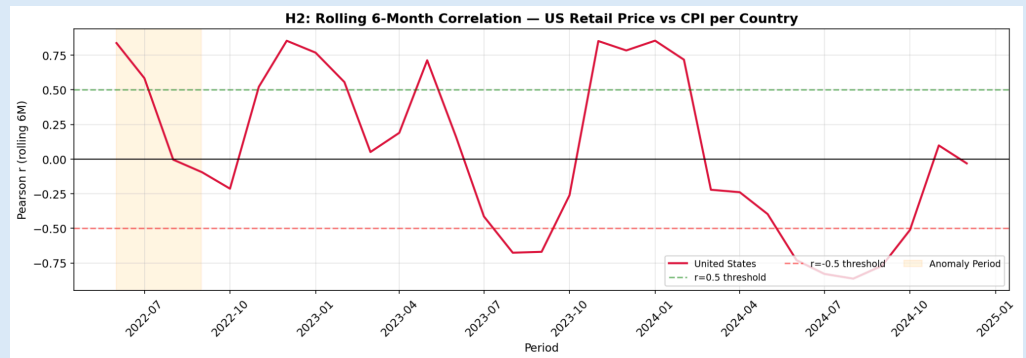


Figure 4. 6-month rolling correlation between U.S. Retail Electricity Prices and domestic CPI

The 6-month rolling Pearson r for the U.S. proves that this relationship is highly volatile. It starts very strong ($r > 0.75$) in early 2022, collapses to zero right after the anomaly period, swings back up in early 2023, drops past the negative threshold ($r < -0.5$) in mid-2023, and spikes again in late 2023. This reveals that the relationship between electricity pricing and CPI is highly dynamic and temporally unstable.

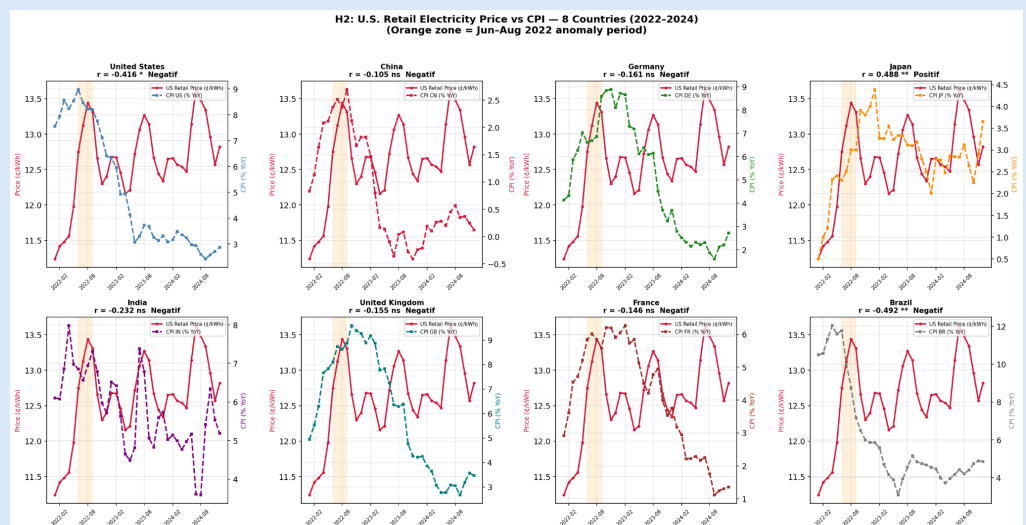


Figure 5. Cross-country comparison of U.S. Retail Electricity prices vs. Consumer Price Index (CPI)

The idea sought to systematically explore the relationship between electricity pricing and Consumer Price Indices (CPI). The charts show a diverse global picture. For the United States, there is a moderate negative correlation ($r = -0.416$). During the June-August 2022 anomaly period, U.S. CPI was peaking globally (around 8-9% y-o-y) while retail electricity prices were on a steep upward trajectory. Countries like Brazil also show a notable negative correlation ($r = -0.492$), whereas Japan stands out with a moderate positive correlation ($r = 0.488$).

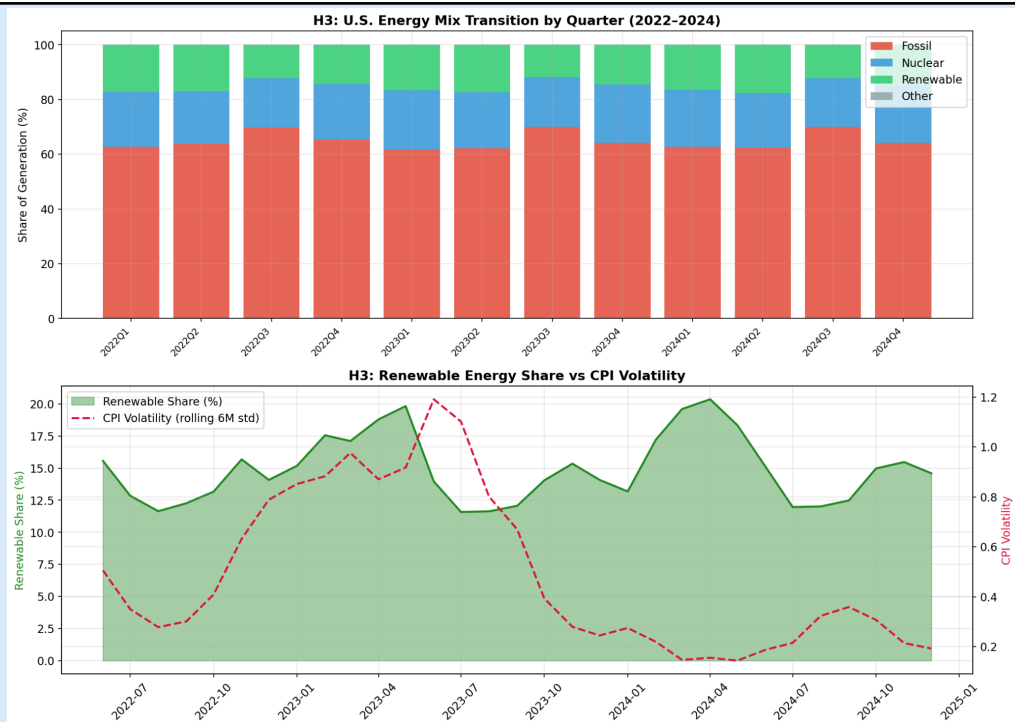


Figure 6. Quarterly U.S. electricity generation mix (top) and renewable energy share vs. CPI volatility trends (bottom)

The stacked bar chart shows that Fossil Fuels consistently dominate U.S. electricity generation, hovering around 60%-70% of the share. Seasonal trends are apparent, with fossil fuel shares consistently peaking in the third quarter (Q3) of 2022, 2023, and 2024 (matching peak summer air conditioning demand).

The renewable energy share experiences a cyclical waveform, dipping in Q3 when fossil fuels peak, but reaching its highest marks (around 20%) in early spring (Q2 2023 and Q2 2024). Interestingly, the rolling 6-month standard deviation of CPI (inflation volatility) peaked significantly in early-to-mid 2023, a period where the renewable energy transition share was rapidly ascending to its local maximum.

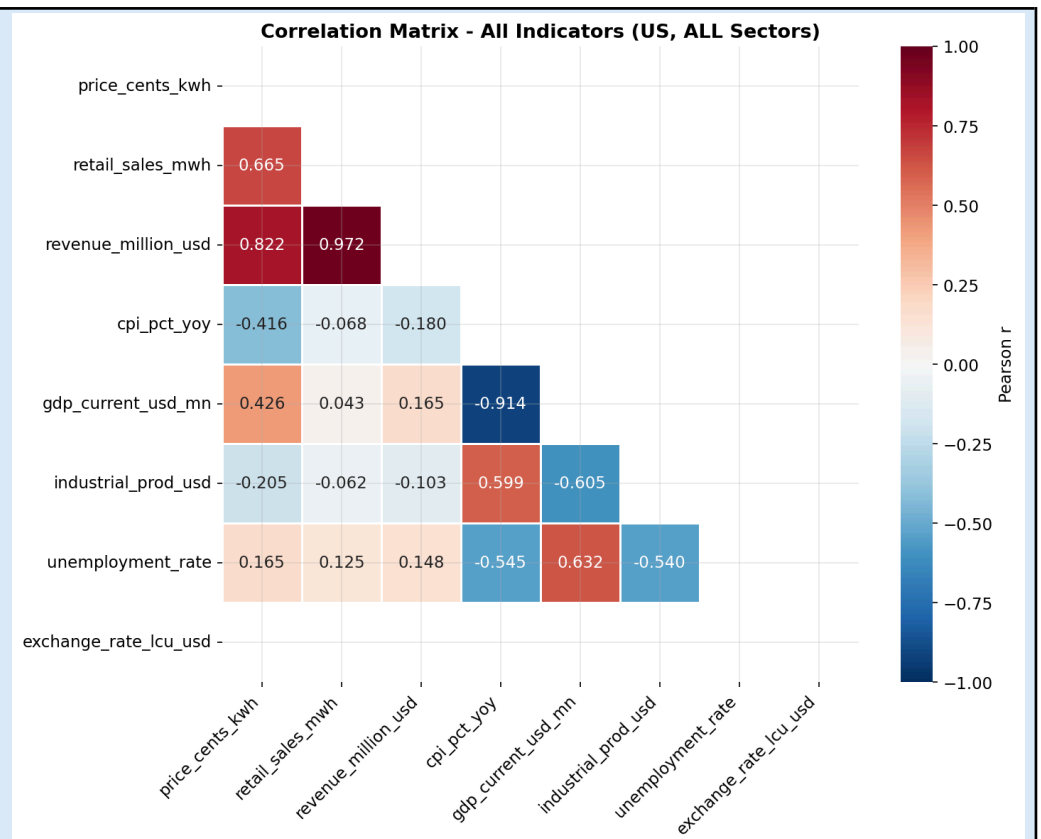


Figure 7. Normalized Z-score heatmap of economic and energy indicators, highlighting the June-August 2022 anomaly period.

Through multidimensional slice-and-dice operations, the June–August 2022 period is clearly isolated as the primary anomaly window, highlighted by a distinct orange border. Within this specific timeframe, the U.S. CPI indicator exhibits a highly intense, deep red saturation, representing the historic annualized inflation peak where its individual Z-score approaches the upper historical boundary ($Z \approx +2.0$). This structural shift closely coincides with a gradual color change in the U.S. Retail Electricity Price indicator moving into the positive spectrum ($Z > +1.0$), demonstrating that intense macroeconomic inflation pressures translated immediately into utility pricing volatility during that quarter.

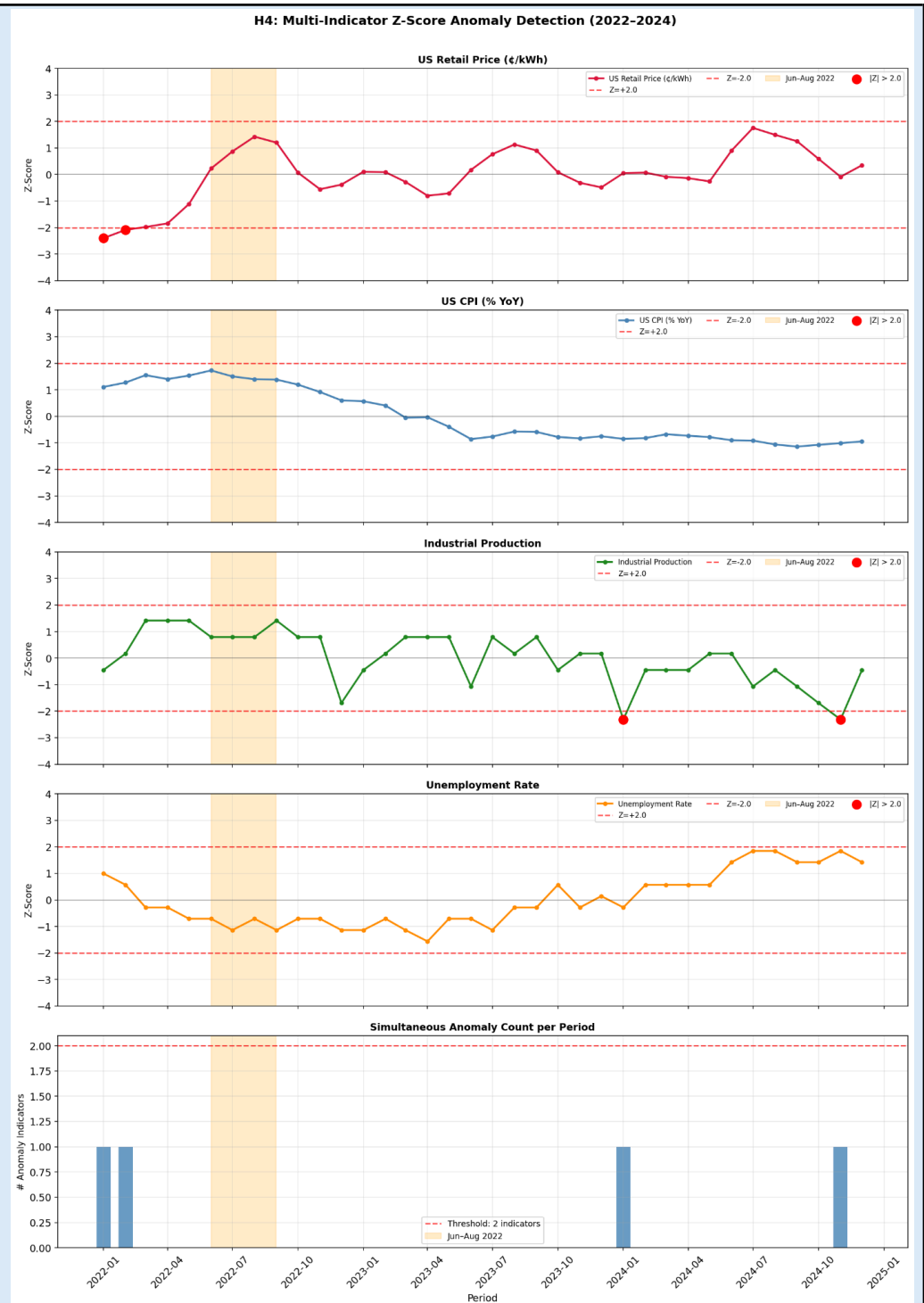


Figure 8. Multivariate anomaly detection and simultaneous indicator breaches based on a $|Z| > 2.0$ threshold framework

Utilizing a multivariate anomaly detection framework within a strict absolute boundary ($|Z| > 2.0$), this line chart tracks the total number of metrics breaching extreme parameters simultaneously within any single given time period. According to the tracking results, although the June-August 2022 window registered immense market volatility, individual indicators (such as CPI and retail Prices) actually breached the strict $|Z| > 2.0$ boundary at slightly shifted, lagging intervals. The total count of concurrent indicator breaches per single month never officially crossed the theoretical anomaly threshold of ≥ 2 variables at once. Instead, isolated extreme anomalies occurred at other independent timestamps, such as a sharp

contraction in the U.S. Industrial Production Index in late 2024 and depressed retail prices in early 2022. This reveals that the transmission of macroeconomic shocks into the energy sector did not manifest as a single synchronized collapse, but rather as a temporally distributed domino effect.

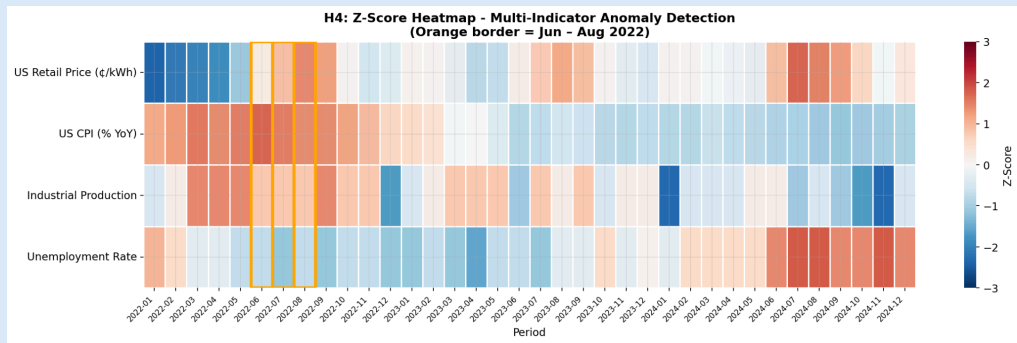


Figure 9. Pearson correlation matrix for all core energy and macroeconomic indicators in the United States

This figure presents a comprehensive Pearson correlation matrix visualizing the pairwise linear relationship among all core energy and macroeconomic indicators within the United States over the 2022-2024 observation period. The matrix employs a color-coded heatmap representation, where red hues indicate positive correlations, blue hues indicate negative correlations, and color intensity reflects correlation strength. Correlation coefficients (r) range from -1.0 (perfect negative linear relationship) to +1.0 (perfect positive linear relationship), with values near zero indicating no linear association